# Chi-Squared Distribution Table

**Chi-square Distribution Table**

| d.f. | .995 | .99 | .975 | .95 | .9 | .1 | .05 | .025 | .01 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 0.00 | 0.00 | 0.00 | 0.00 | 0.02 | 2.71 | 3.84 | 5.02 | 6.63 |
| 2 | 0.01 | 0.02 | 0.05 | 0.10 | 0.21 | 4.61 | 5.99 | 7.38 | 9.21 |
| 3 | 0.07 | 0.11 | 0.22 | 0.35 | 0.58 | 6.25 | 7.81 | 9.35 | 11.34 |
| 4 | 0.21 | 0.30 | 0.48 | 0.71 | 1.06 | 7.78 | 9.49 | 11.14 | 13.28 |
| 5 | 0.41 | 0.55 | 0.83 | 1.15 | 1.61 | 9.24 | 11.07 | 12.83 | 15.09 |
| 6 | 0.68 | 0.87 | 1.24 | 1.64 | 2.20 | 10.64 | 12.59 | 14.45 | 16.81 |
| 7 | 0.99 | 1.24 | 1.69 | 2.17 | 2.83 | 12.02 | 14.07 | 16.01 | 18.48 |
| 8 | 1.34 | 1.65 | 2.18 | 2.73 | 3.49 | 13.36 | 15.51 | 17.53 | 20.09 |
| 9 | 1.73 | 2.09 | 2.70 | 3.33 | 4.17 | 14.68 | 16.92 | 19.02 | 21.67 |
| 10 | 2.16 | 2.56 | 3.25 | 3.94 | 4.87 | 15.99 | 18.31 | 20.48 | 23.21 |

Degrees of freedom = number of parameters that are free to vary. $p + q = 1$, so $df$=1.

# Chi-Squared Distribution

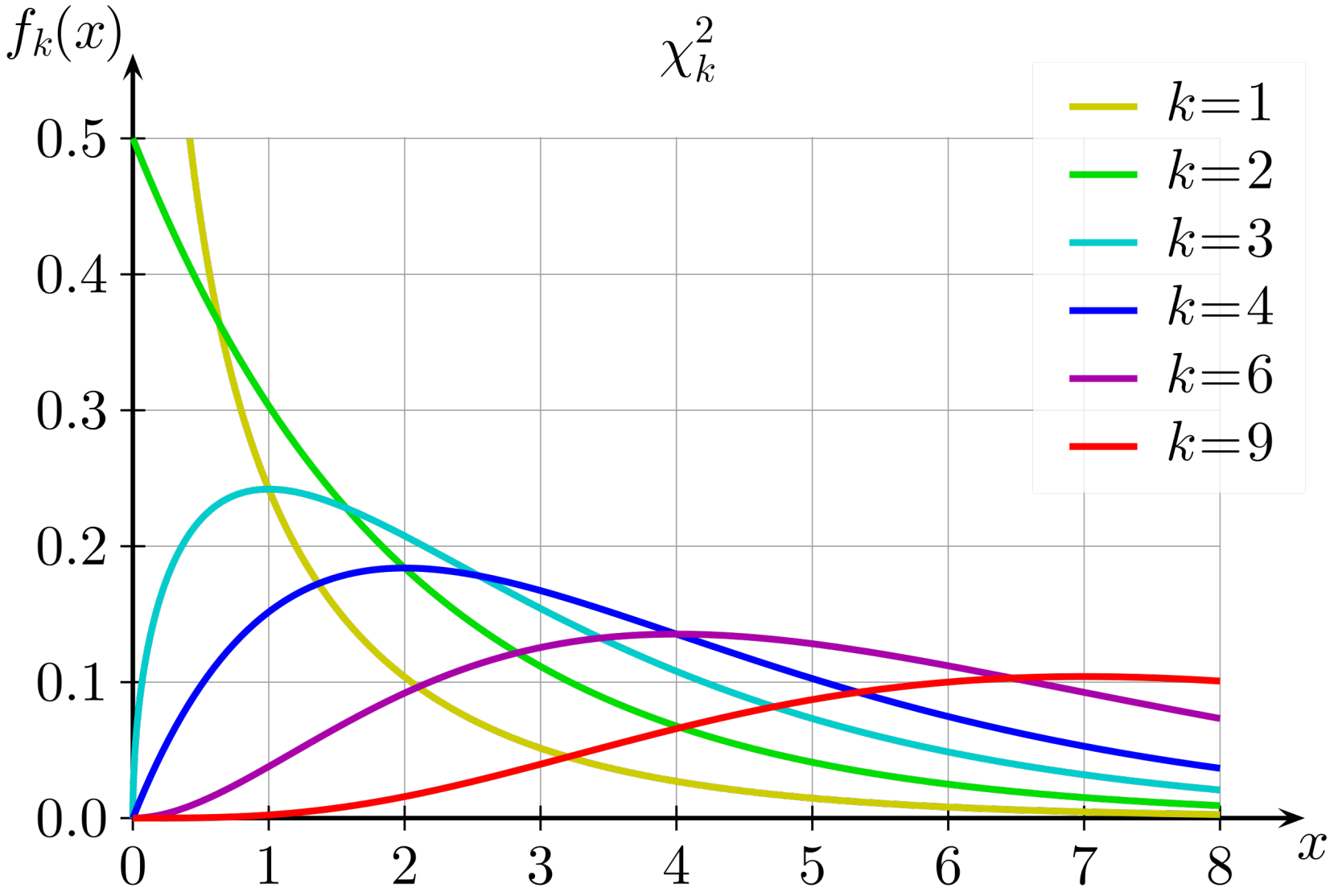
Probability Density Function:

$$f(k) = \frac{1}{2^{k/2}\Gamma(k/2)} x^{k/2-1} e^{-x/2}$$

where $k$ = degrees of freedom and $x$ is the test statistic (i.e., what you use to determine the p-value)

# Chi-Squared Distribution

Probability Density Function:



# Ferruginous Hawks

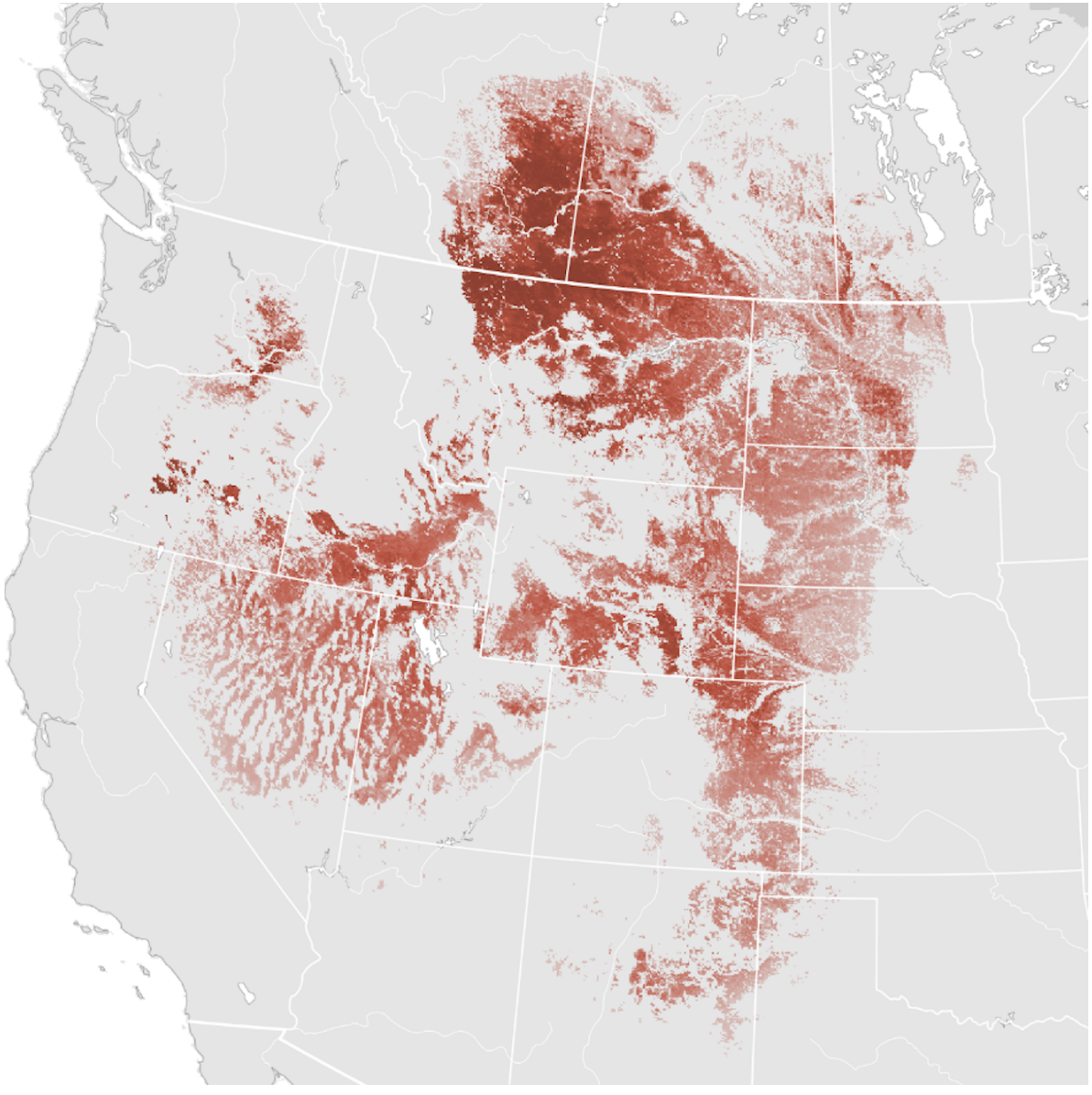


light morph



dark morph

# Ferruginous Hawks



# Ferruginous Hawks



genotype aa



genotypes Aa, AA

# Discussion Questions (Optional Reading)

1) Schmutz & Schmutz 1981: Why might there have been a single dark nestling observed in the light x light mating events in 1977?

2) Schmutz & Schmutz 1981: What does Table 2 tell us?